

CSE 311:

Data Communication

Instructor:
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Analysis of Signals and Systems

Properties of Fourier Transform

Convolution and
Modulation Property

Time Convolution

$$g_1(t) * g_2(t) \Leftrightarrow G_1(f)G_2(f)$$

Modulation or frequency Convolution

$$g_1(t)g_2(t) \Leftrightarrow G_1(f) * G_2(f)$$

Review

Properties of Fourier Transform

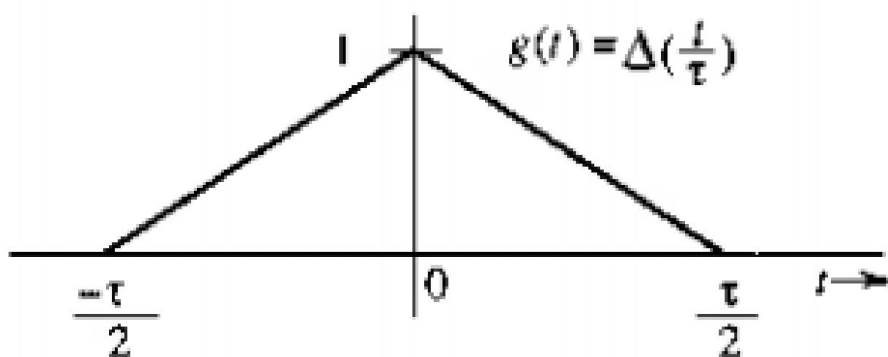
What is Convolution?

$$g_1(t) * g_2(t) = \int_{-\infty}^{\infty} g_1(\tau) g_2(t - \tau) d\tau$$

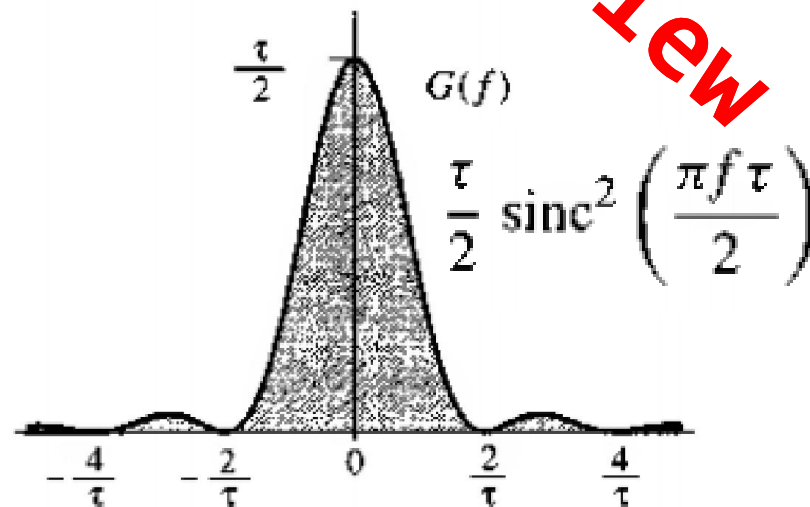
Review

Time Domain and Frequency Domain Representation

Review



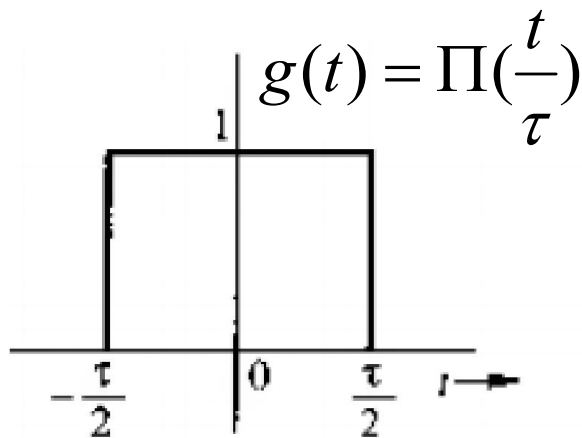
Limited in time



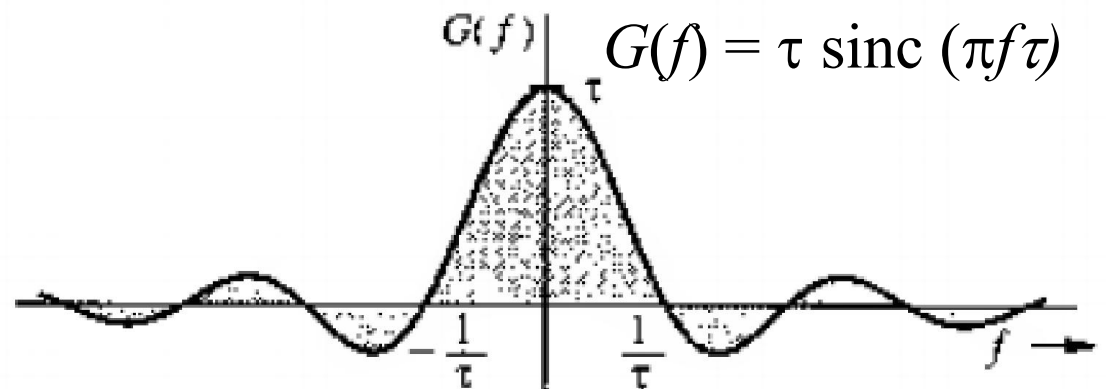
Indefinite spread in frequency

Time Domain and Frequency Domain Representation

Review



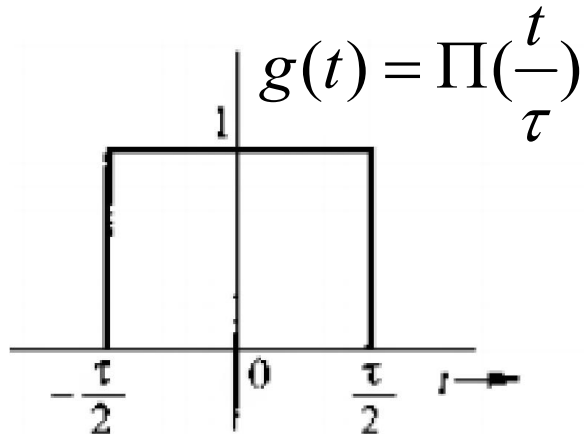
Limited in time



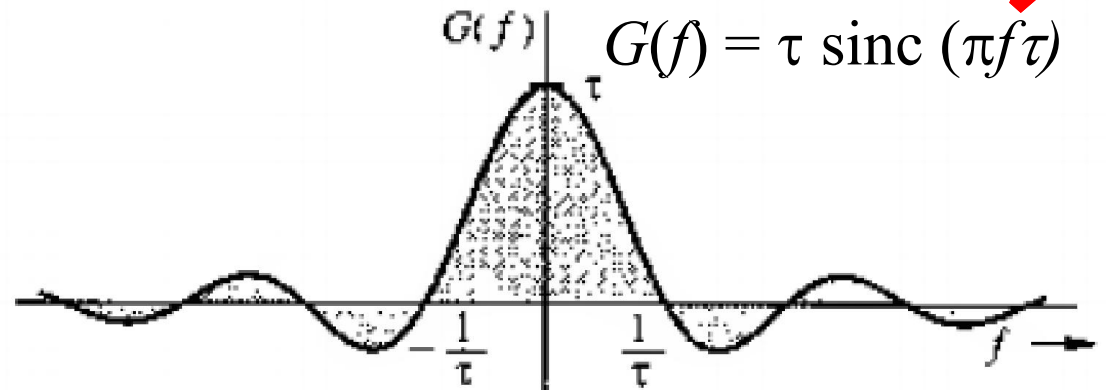
Indefinite spread in frequency

Time Domain and Frequency Domain Representation

Review

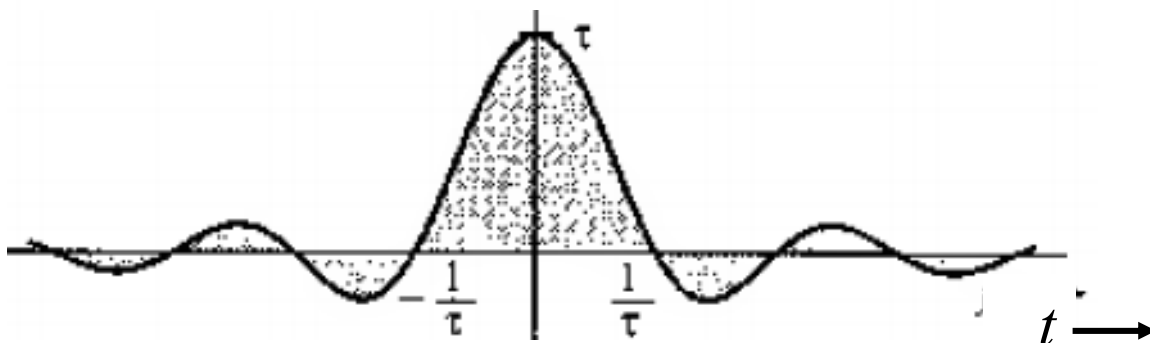


Limited in time

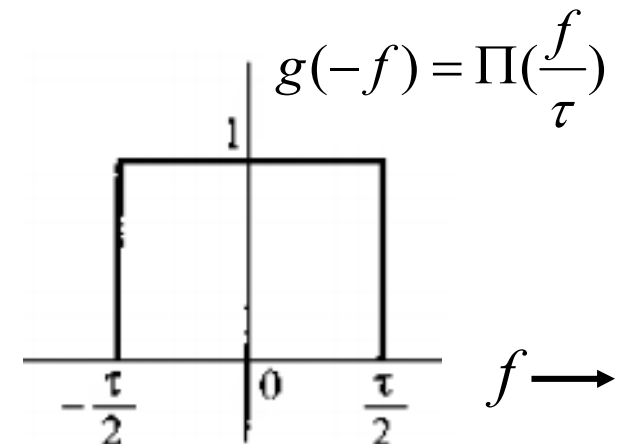


Indefinite spread in frequency

$$G(t) = \tau \text{sinc}(\pi \tau t)$$



Indefinite spread in time



Limited in frequency

Time Domain and Frequency Domain Representation

Review

- A signal cannot be BOTH band limited and time limited
- Relation is inverse
- Relation changes in inverse manner
- Any change in time specification changes spectrum specification inversely
- We can specify EITHER of them NOT BOTH of them

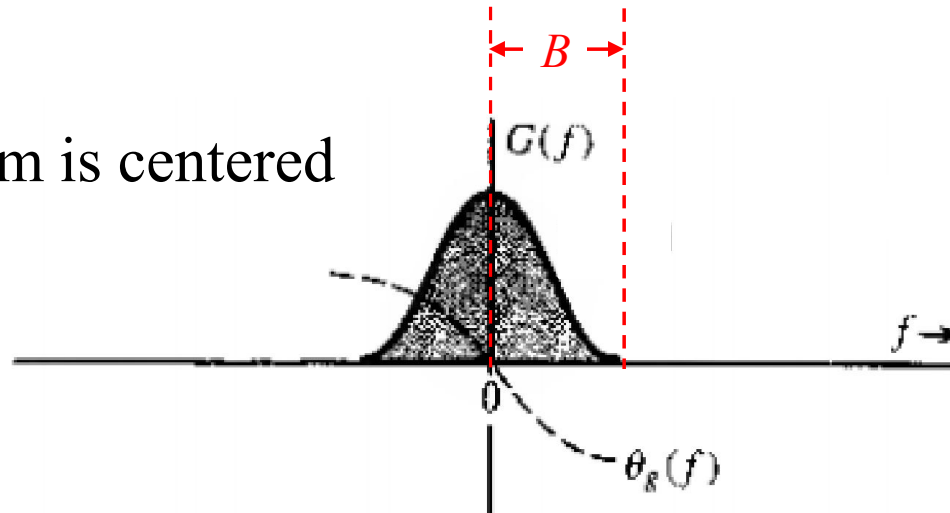
Bandwidth of a Signal

- Definition: extent of SIGNIFICANT spectral content of a signal for POSITIVE frequencies
- Alternate definitions exist

Bandwidth of a Signal

Lowpass signal:

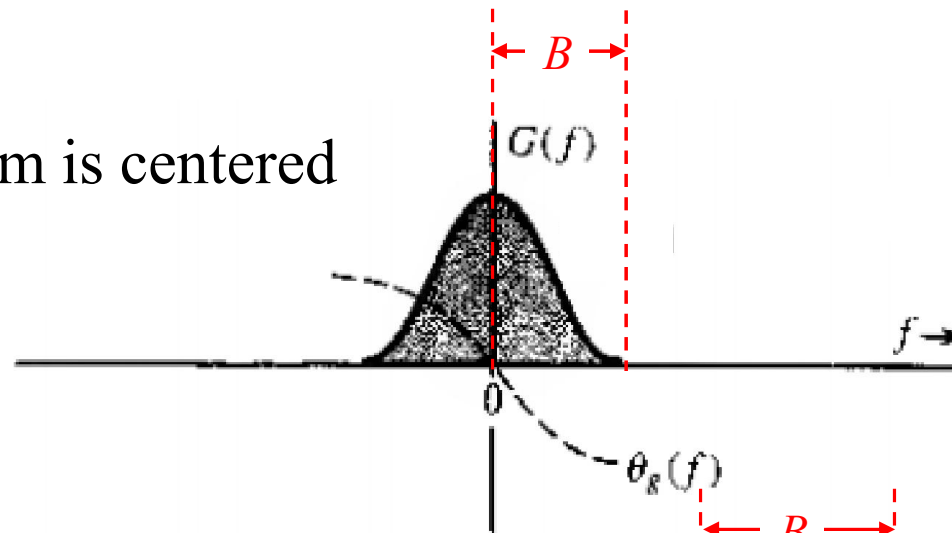
Frequency spectrum is centered around $f = 0$



Bandwidth of a Signal

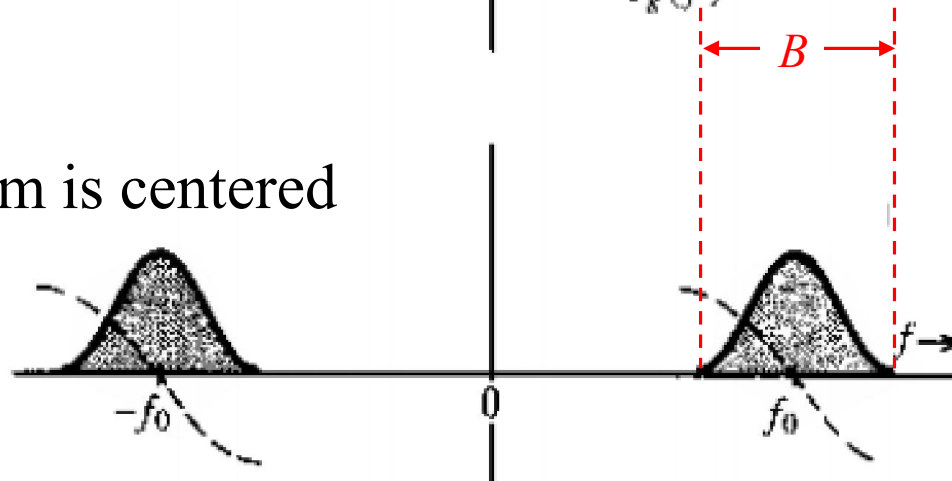
Lowpass signal:

Frequency spectrum is centered around $f = 0$



Bandpass signal:

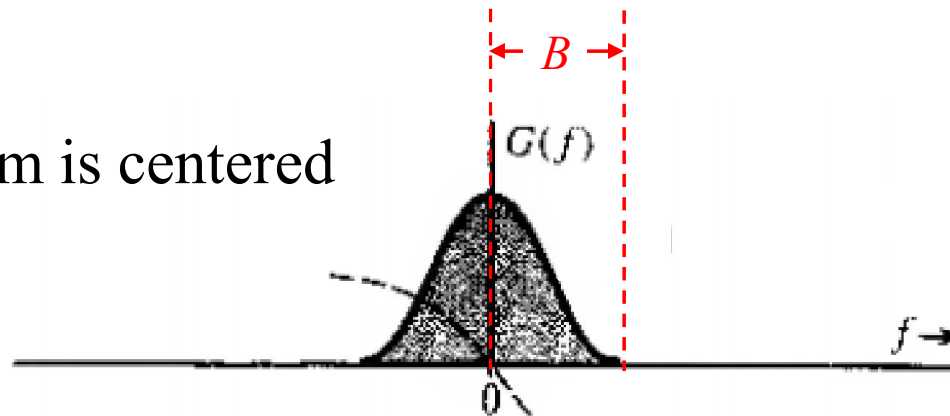
Frequency spectrum is centered around $\pm f_0$



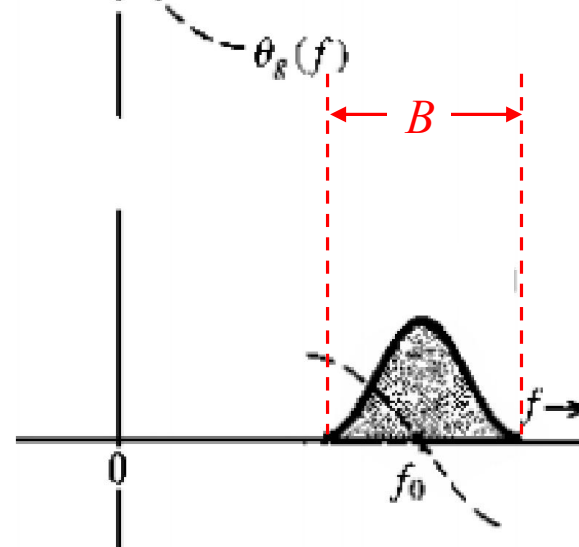
Bandwidth of a Signal

Lowpass signal:

Frequency spectrum is centered around $f = 0$

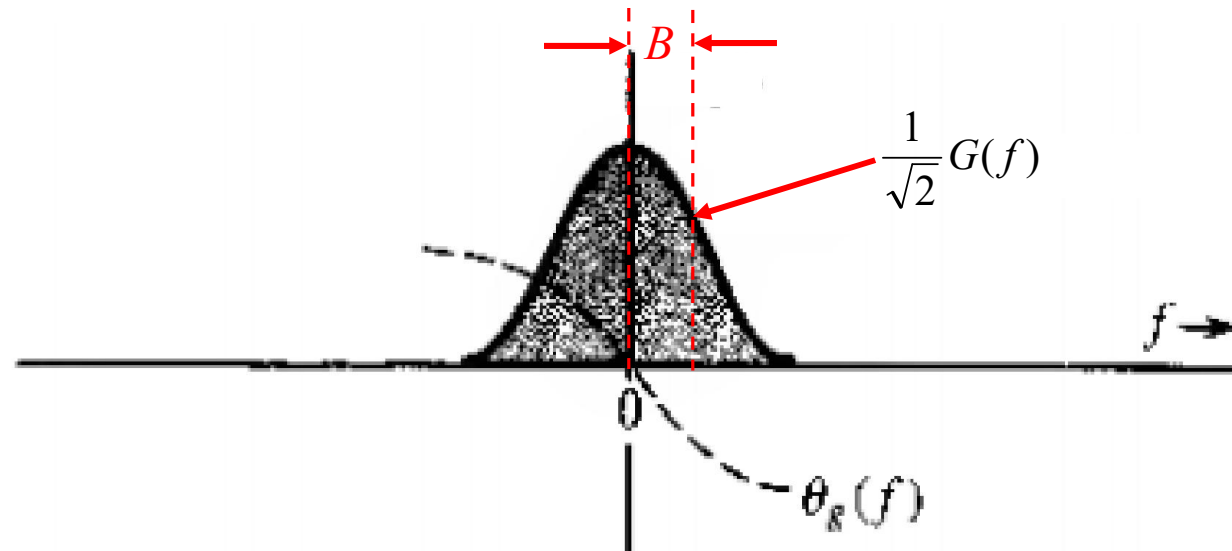


Shifting a low pass signal to f_0
location **DOUBLES** its
bandwidth



3-dB Bandwidth of a Signal

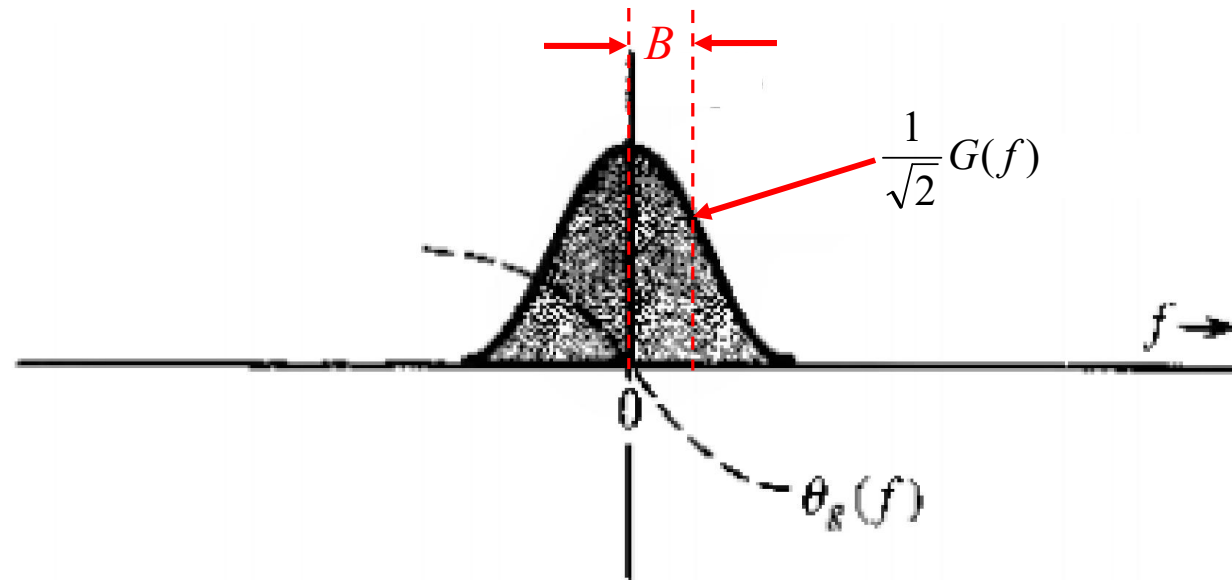
Lowpass signal:
Separation betn
peak and the
position where it
is $\frac{1}{\sqrt{2}}$ of its peak



3-dB Bandwidth of a Signal

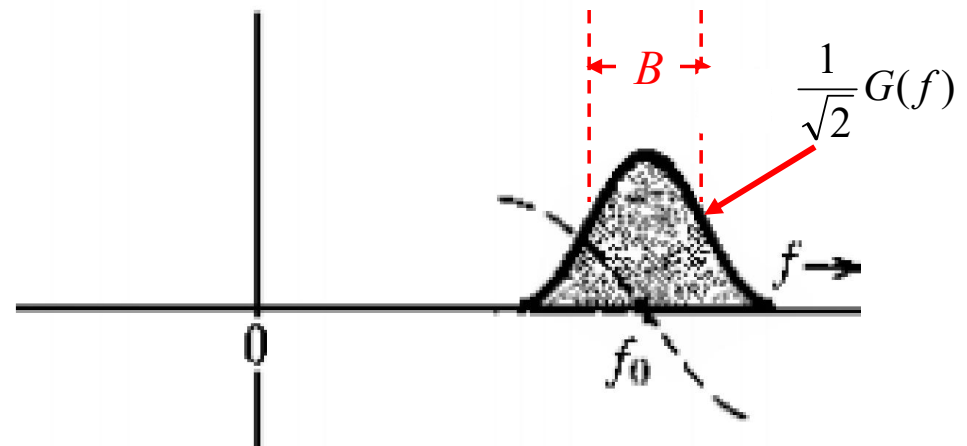
Lowpass signal:

Separation betn
peak and the
position where it
is $\frac{1}{\sqrt{2}}$ of its peak



Bandpass signal:

Separation betn two positions
where amplitudes are $\frac{1}{\sqrt{2}}$ of its
peak



rms Bandwidth of a Signal

Root mean square bandwidth:

the square root of the second moment of a properly normalized form of the squared amplitude spectrum of the signal about a suitably chosen point

rms Bandwidth of a Signal

Root mean square bandwidth:

the square root of the second moment of a properly normalized form of the squared amplitude spectrum of the signal about a suitably chosen point

k^{th} -moment of squared amplitude about origin

$$= \int_{-\infty}^{\infty} f^k |G(f)|^2 df$$

rms Bandwidth of a Signal

Root mean square bandwidth:

the **square root** of the **second moment** of a properly **normalized form** of the **squared amplitude** spectrum of the signal **about a suitably chosen point**

2nd moment of squared amplitude about origin

$$m_2 = \int_{-\infty}^{\infty} f^2 |G(f)|^2 df$$

0-th moment of squared amplitude about origin

$$m_0 = \int_{-\infty}^{\infty} |G(f)|^2 df$$

rms Bandwidth of a Signal

Root mean square bandwidth:

the square root of the second moment of a properly normalized form of the squared amplitude spectrum of the signal about a suitably chosen point

2nd moment of normalized squared amplitude about origin

$$= \frac{\int_{-\infty}^{\infty} f^2 |G(f)|^2 df}{\int_{-\infty}^{\infty} |G(f)|^2 df} = \frac{m_2}{m_0}$$

rms Bandwidth of a Signal

Root mean square bandwidth:

the square root of the second moment of a properly normalized form of the squared amplitude spectrum of the signal about a suitably chosen point

$$rms \text{ bandwidth, } W_{rms} = \sqrt{\frac{\int_{-\infty}^{\infty} f^2 |G(f)|^2 df}{\int_{-\infty}^{\infty} |G(f)|^2 df}} = \sqrt{\frac{m_2}{m_0}}$$

Time-Bandwidth Product

$$\text{Duration} \times \text{bandwidth} = \text{constant}$$

- Relationship is true for every definition of bandwidth

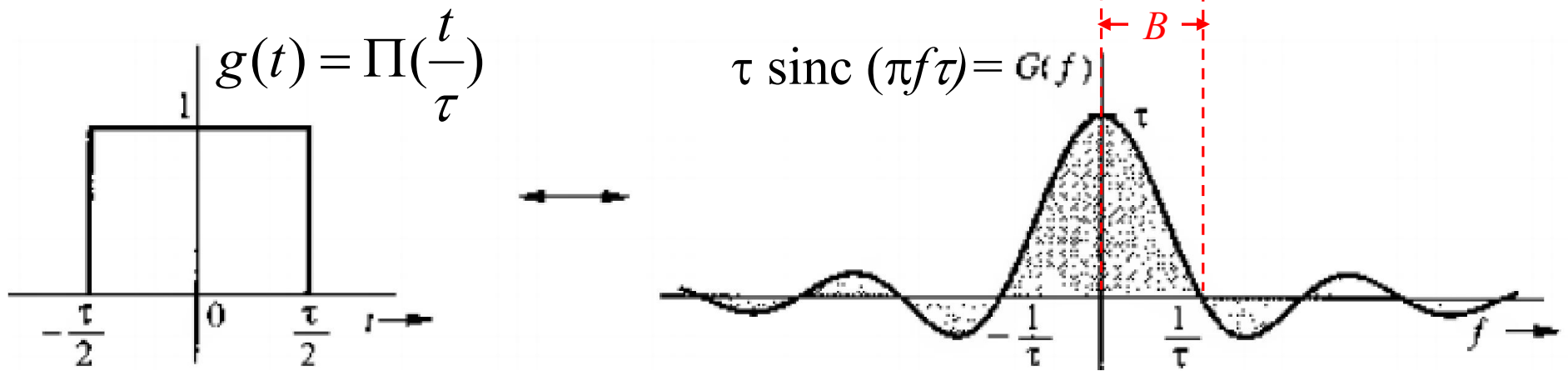
Time-Bandwidth Product

$$\text{Duration} \times \text{bandwidth} = \text{constant}$$

- Relationship is true for every definition of bandwidth
- Compression in time by a factor a expands the spectrum by the same factor and vice versa
 - No change in the constant

Time-Bandwidth Product

Duration $\times$ bandwidth = constant

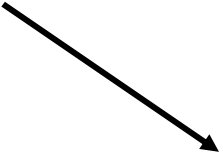


Duration $\times$ bandwidth = $\tau \times (1/\tau) = 1 = \text{constant}$

Time-Bandwidth Product

Duration $\times$ bandwidth = constant


$$T_{rms} = \sqrt{\frac{\int_{-\infty}^{\infty} t^2 |g(t)|^2 dt}{\int_{-\infty}^{\infty} |g(t)|^2 dt}}$$


$$W_{rms} = \sqrt{\frac{\int_{-\infty}^{\infty} f^2 |G(f)|^2 df}{\int_{-\infty}^{\infty} |G(f)|^2 df}}$$

Time-Bandwidth Product

Duration $\times$ bandwidth = constant

$$T_{rms} \times w_{rms} \geq \frac{1}{4\pi}$$

Time-Bandwidth Product

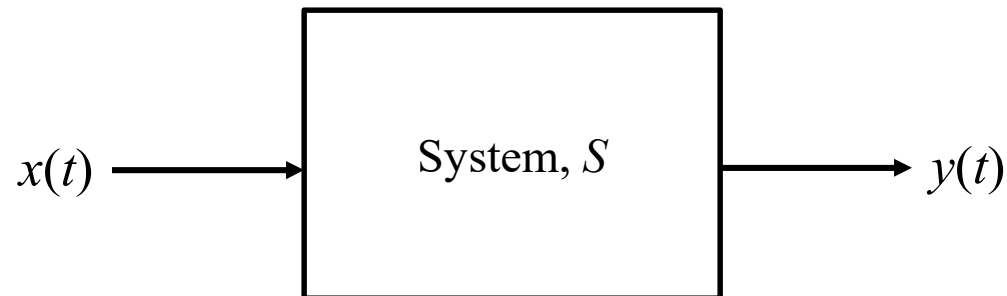
Duration $\times$ bandwidth = constant

$$T_{rms} \times w_{rms} \geq \frac{1}{4\pi}$$

- Equality holds for Gaussian pulse

Signal Transmission through Linear Time Invariant System

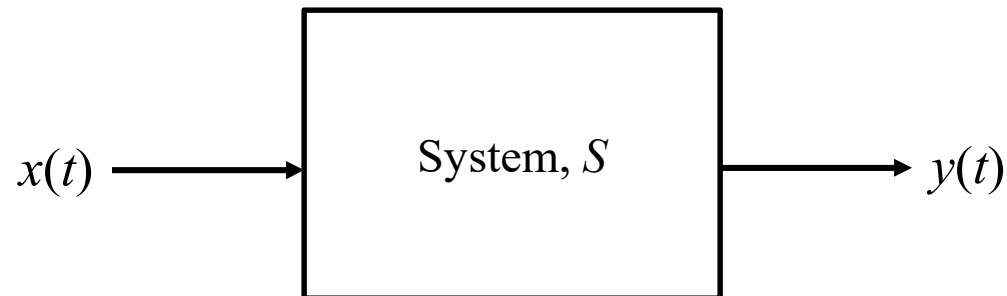
System



$$y(t) = S(x(t))$$

Signal Transmission through Linear Time Invariant System

System



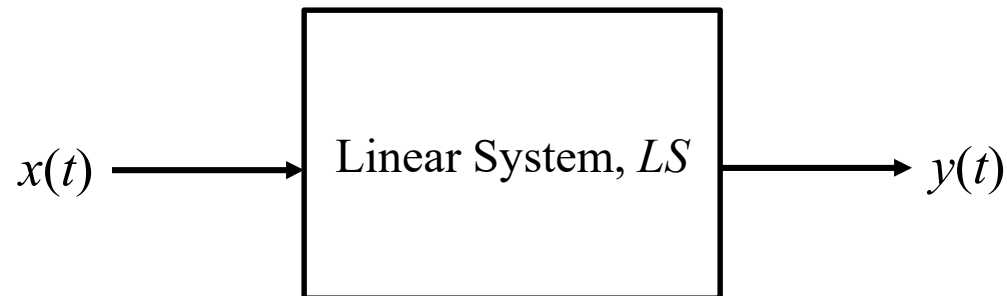
$$y(t) = S(x(t))$$

$$y_1(t) = S(x_1(t))$$

$$y_2(t) = S(x_2(t))$$

Signal Transmission through Linear Time Invariant System

Linear System

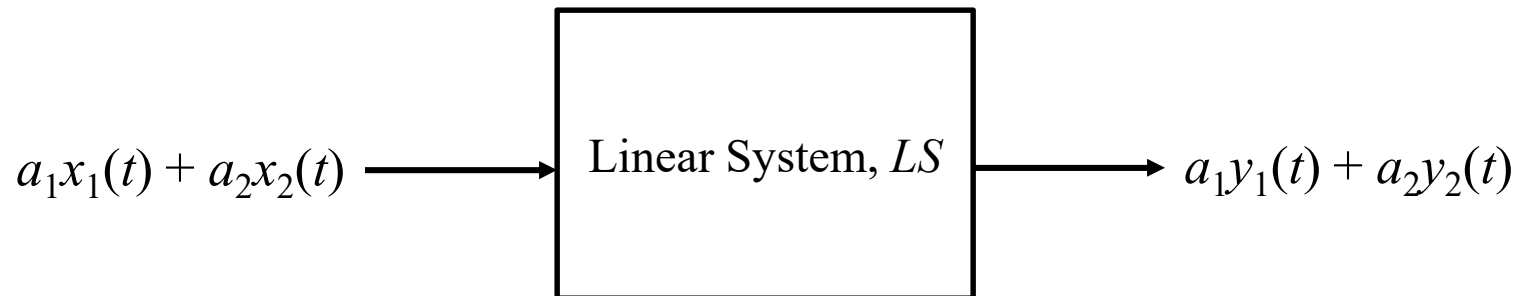


$$y(t) = LS(x(t))$$

$$y_1(t) = LS(x_1(t)) \qquad y_2(t) = LS(x_2(t))$$

Signal Transmission through Linear Time Invariant System

Linear System



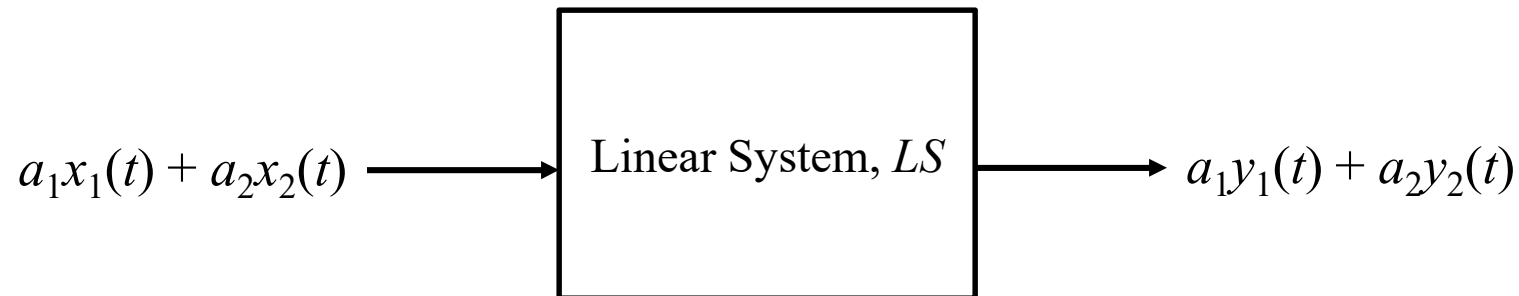
$$y(t) = LS(x(t))$$

$$y_1(t) = LS(x_1(t)) \qquad y_2(t) = LS(x_2(t))$$

$$a_1y_1(t) + a_2y_2(t) = LS(a_1x_1(t) + a_2x_2(t))$$

Signal Transmission through Linear Time Invariant System

Linear System

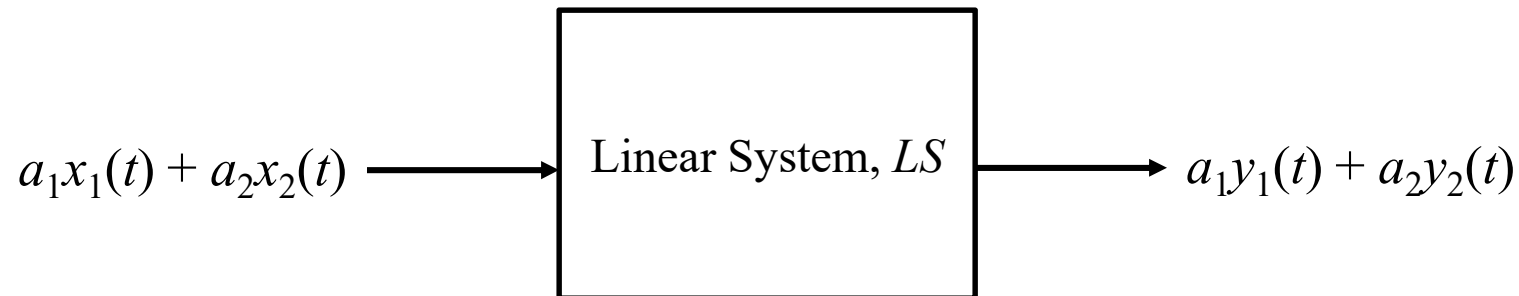


Principle of Superposition

the response of a linear system to a number of excitations applied simultaneously is equal to the sum of the responses of the system when each excitation is applied individually.

Signal Transmission through Linear Time Invariant System

Linear System



Principle of Superposition

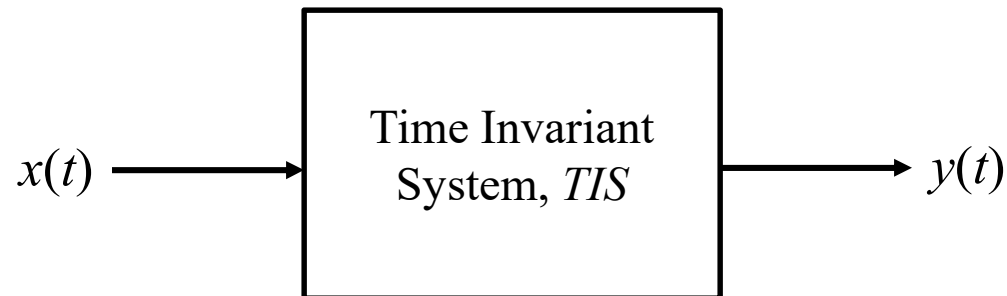
the response of a linear system to a number of excitations applied simultaneously is equal to the sum of the responses of the system when each excitation is applied individually.

Examples

Filters and Communication Channels

Signal Transmission through Linear Time Invariant System

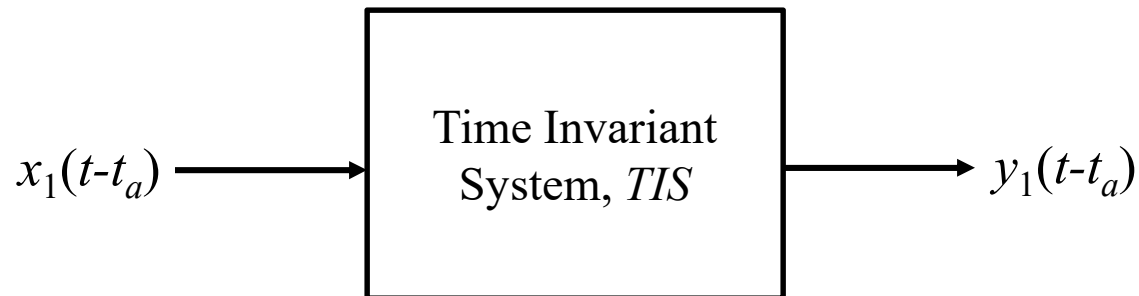
Time Invariant System



$$y(t) = TIS(x(t))$$

Signal Transmission through Linear Time Invariant System

Time Invariant System

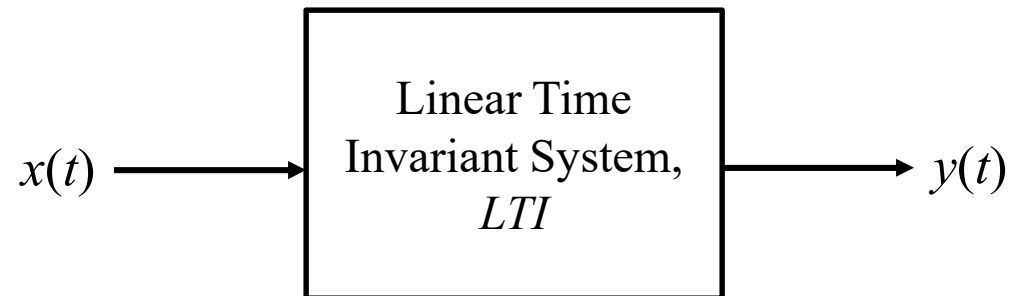


$$y(t) = TIS(x(t))$$

$$y_1(t-t_a) = TIS(x_1(t-t_a)) \quad y_2(t-t_b) = TIS(x_2(t-t_b))$$

Signal Transmission through Linear Time Invariant System

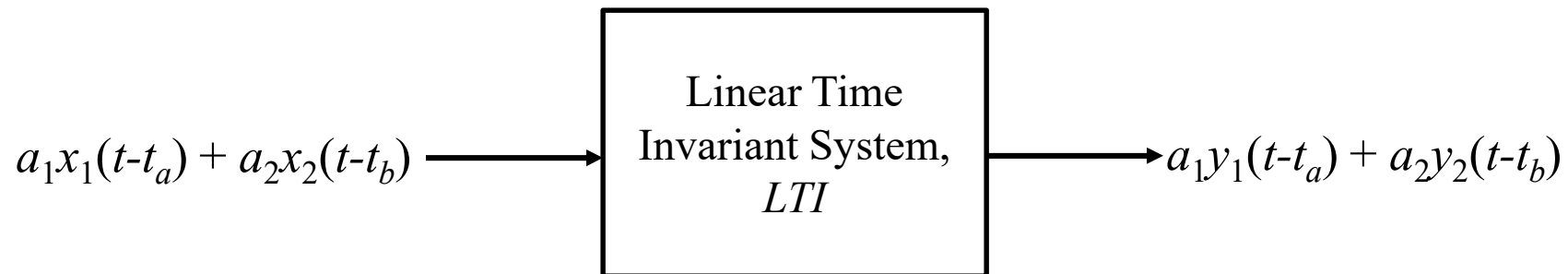
Linear Time Invariant System



$$y(t) = LTI(x(t))$$

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System



$$y(t) = LTI(x(t))$$

$$a_1y_1(t-t_a) + a_2y_2(t-t_b) = LTI(a_1x_1(t-t_a) + a_2x_2(t-t_b))$$